

STEP 1: DEFINING THE DATASETS

The final master dataset contains 72 monthly observations from January 2019 to December 2024 across 14 variables. We used a mix of macroeconomic variables, financial, as well as oil market metrics. We also added derived values such as WTI return, volatility, etc. After we cleaned the data, we divided it into three different parts.

Table 1. Dataset Allocation Summary

Dataset	Purpose	Start Date	End Date	Observations	Share of Total Data (%)
Training Set	Model learning, parameter estimation, regime detection, and network structure learning	2019-01-31	2023-09-30	57	79.1700
Validation Set	Model selection, tuning, and checking whether the model generalizes beyond training	2023-10-31	2024-04-30	7	9.7200
Testing Set	Final out-of-sample evaluation of forecasting accuracy and model reliability	2024-05-31	2024-12-31	8	11.1100

The data is divided into 57 monthly observations for the purpose of training(which is 79%). The validation set is 7(9.7%) and the set we used for testing is 8(11%). We make sure to obey the 80:10:10 model of allocation.

Student A: The Purpose of the Training Set

Think of the training set as OUR model’s memory, it’s where the actual learning happens. This dataset covers January 2019 to September 2023, which makes it the largest and earliest chunk of our data. We chose this specific timeframe because it capture a wild mix of market conditions such as normal

months, the massive COVID-19 demand crash, the subsequent recovery, the Russia-Ukraine energy shock, and also the recent cycle of inflation and rising interest rates.

For crude oil forecasting, having this diverse history is very important. The price of WTI crude isn't driven by one factor. It reacts to a whole lot of forces, including demand, production, inventories, inflation, interest rates, the U.S. dollar, and others. If we built a model based purely on historical prices, we will miss all these important connections. Giving our Bayesian network this broad training set gives us a realistic and grounded foundation for understanding different market states. Basically, the training set builds the structure and maps out the relationships. However, just because a model perfectly explains the past doesn't mean it can handle the future. That's exactly why we keep the validation and testing sets entirely separate.

Student B: The Purpose of the Validation Set

The validation set acts as our first real quality-control checkpoint. Since it covers October 2023 to April 2024, this data isn't used to teach the model its core rules. Instead, it lets us see if a model trained on older data can still make logical and economic sense when thrown into a new timeframe it hasn't seen before.

The rules of the oil market are always changing. The way things like interest rates, inflation, inventories, the U.S. dollar, and volatility affect crude oil changes, but depending on the overall economic environment [Alvi, 2018]. The validation set gives us a way to verify the model before the final exam. There is one important rule here, though: we can't keep tweaking the model until it perfectly fits the validation data. If we do that, the validation set essentially just becomes a second training set, which, will most likely, completely ruin the honesty of our final test.

Student C: The Purpose of the Testing Set

The testing set is our final and untouched holdout sample, which runs from May 2024 to December 2024. We only touch this data after the model is completely trained and validated. Its sole purpose is to answer one question: can our finished model actually produce reliable predictions using data it has never seen or learned from? This part is the closest we get to the real world. Traders, portfolio managers, and risk analysts don't care about a model that is only good at explaining what happened yesterday. They need a tool that helps them make decisions when new monthly data is available. The testing set is that ultimate, out-of-sample proof that the model actually works.

STEP 2: COMPARING THE VALIDATION AND TESTING SETS AND SHOWING DATA ALLOCATION

As the charts show, we split our data chronologically: the training period comes first, the validation period follows, and then the testing period. We made sure that these three timeframes never

overlap. This separation we did was important because it protects the model from "look-ahead bias", this means, the model never gets to cheat by peeking at future data.

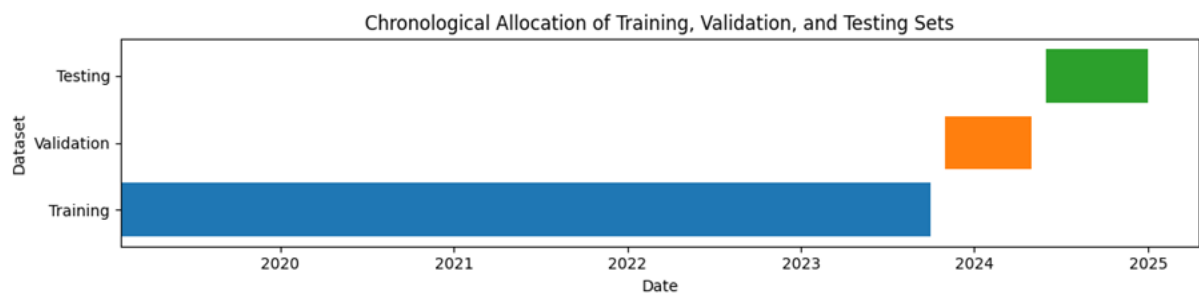


Figure 1. Chronological allocation of training, validation, and testing sets.

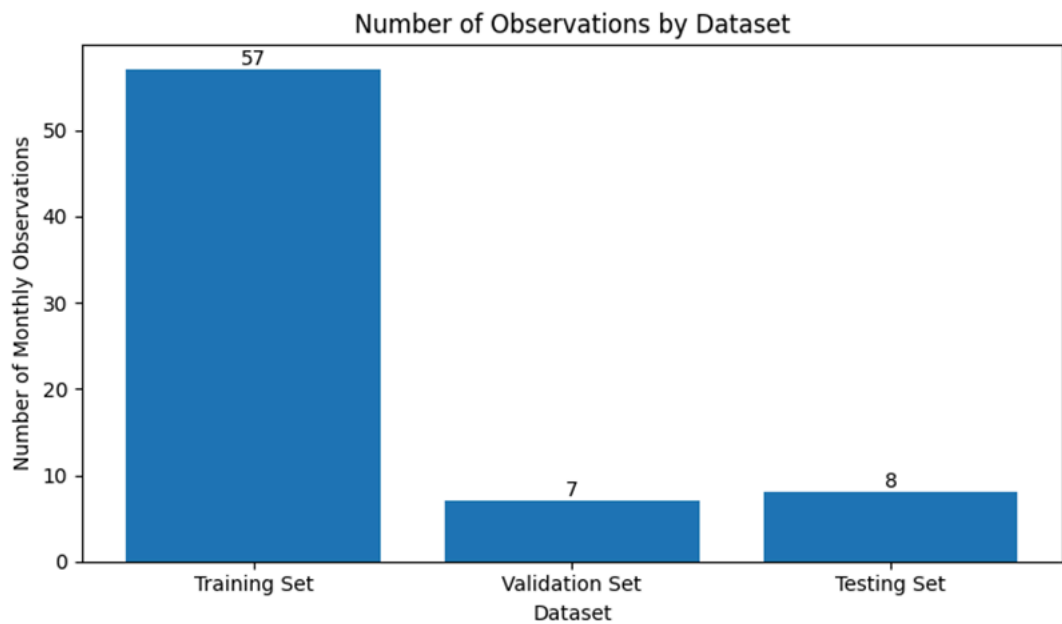


Figure 2. Number of monthly observations in each dataset.

By design, the largest chunk of our data was used for the training phase. This ensured that the model had enough history to learn how the market behaves under various conditions. Meanwhile, keeping the validation and testing periods entirely separate ensures that our final assessment of the model is completely fair.

When we looked at the validation and testing periods, we realized they weren't identical. In fact, our statistical checks flagged potential shifts in 10 out of our 14 variables. Hence our final testing period wasn't just a copy-paste of the validation phase, the model was being tested in a market environment that had evolved and changed.

Table 2. Validation versus Testing Comparison

Variable	Validation Mean	Testing Mean	Mean Difference	Standardized Mean Difference	Drift Flag
Interest Rate	5.3300	5.0500	-0.2800	-1.1288	Possible shift
CPI	310.0883	314.7931	4.7048	2.5000	Possible shift
Industrial Production	100.3063	100.1327	-0.1736	-0.3092	No major shift
USD Index	121.1601	124.2768	3.1167	1.3829	Possible shift
Production	13139.4286	13333.1250	193.6964	0.8819	Possible shift
Inventory	440452.5714	426795.2500	-13657.3214	-0.9507	Possible shift
Demand	20241.8571	20698.7500	456.8929	1.3599	Possible shift
WTI	78.2629	73.3925	-4.8704	-1.0653	Possible shift
Brent	83.9929	77.5075	-6.4854	-1.4044	Possible shift
VIX	14.2743	15.9338	1.6595	0.5880	No major shift
Energy ETF	40.6108	43.1044	2.4936	1.2733	Possible shift
WTI Return	-0.0147	-0.0166	-0.0020	-0.0325	No major shift
Volatility	0.0594	0.0496	-0.0099	-0.3713	No major shift
Spread	5.7300	4.1150	-1.6150	-1.8233	Possible shift

Here is what that change looked like: During the testing period, crude prices were low. The average price of WTI crude fell from \$78.26 in validation to \$73.39 in testing, and Brent crude dropped from

\$83.99 to \$77.51. At the exact same time, inflation (CPI) was higher, the U.S. dollar was strong, demand and production had increased, and inventories were lower.

In essence, the testing period wasn't weak because the fundamentals collapsed; rather, it was the crude prices that softened because heavy financial and macroeconomic pressures were on the market.

Because of this, if the model struggled during the final testing phase; hence we shouldn't assume it's a technical failure. It might simply mean that the underlying market environment shifted. When forecasting crude oil, this is completely normal. Oil prices are pushed and pulled by many different forces at once, and the strength of those relationships is always changing over time [Alvi, 2018].

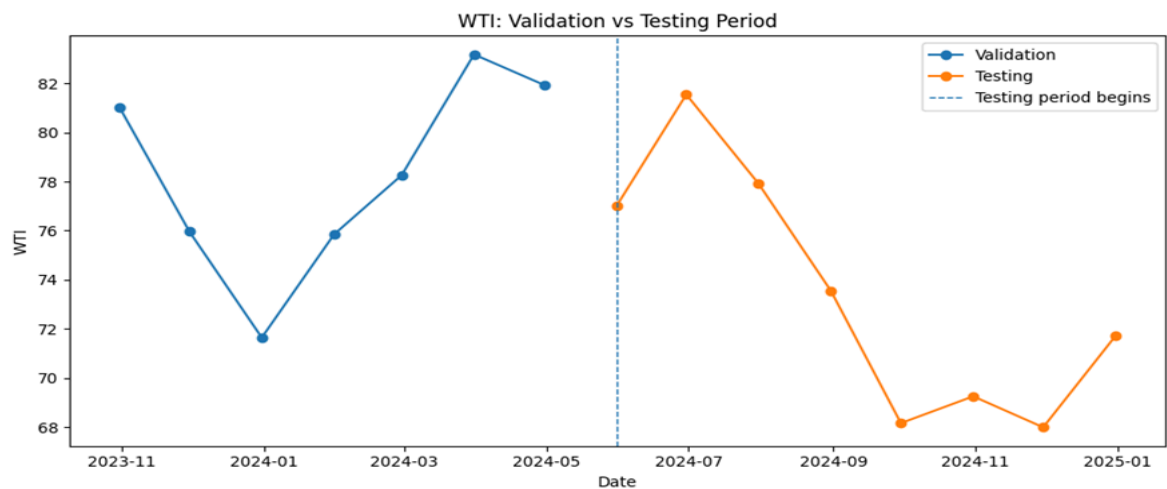


Figure 3. WTI price behaviour across the validation and testing periods.

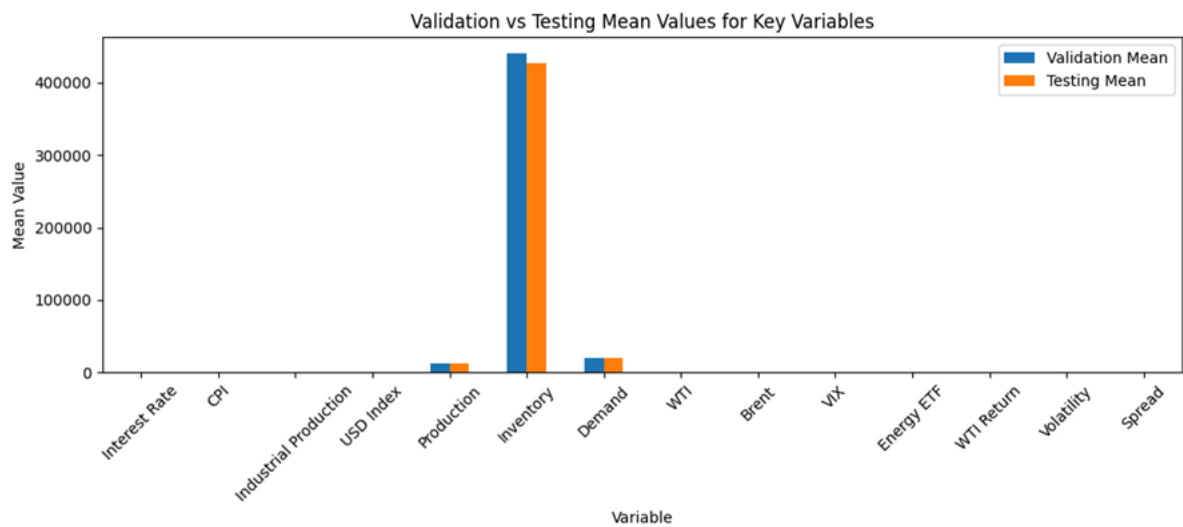


Figure 4. Mean values of key variables in the validation and testing period

STEP 3: MODEL VALIDATION FRAMEWORK

Alvi’s approach to validation compares the model's predictions against what happened in the market and then calculates an error rate. It is a way to see if our forecasted market states match reality. As a bonus, the comparison makes it easy to explain our results to non-technical readers and/or decision-makers.

For our project, we built our validation framework around the rules listed below:

- 1. Keep the timelines separate, that is, our training, validation, and testing data must be divided by time.
- 2. There must be no cheating on the final test: We absolutely cannot use the final testing set to tune the model.
- 3. Consider the fact that context is everything: We judge the results using statistical math and real-world economics.

STEP 4: RE-RUNNING THE BAYESIAN NETWORK USING HILL CLIMBING

We used a hill-climbing algorithm with a BIC scoring system to map out how our variables connect. The final score is -827.94, and it revealed 12 different data connections. Interestingly, the algorithm didn't find a direct, and single path leading to our forecast target. To solve this, we used a statistical method called "mutual information" to manually identify the most powerful predictors. The two clear winners were the VIX and Interest Rates.. Market volatility (the VIX) and the broad monetary environment (interest rates) are key factors when it comes to understanding and pricing risk in the crude oil market.

Table 4. Learned Bayesian Network Edges

Parent	Child
Brent	WTI
CPI	Energy ETF
CPI	Interest Rate
CPI	USD Index
Demand	Industrial Production
Energy ETF	Brent
Industrial Production	CPI
Industrial Production	Inventory
Interest Rate	Production
Interest Rate	WTI Return
Production	Spread

VIX	Demand
-----	--------

The connections our model finds includes a linked Brent to WTI, which is what you'd expect, since they are the world's primary crude oil benchmarks. It also connected inflation (CPI) to the energy ETF, interest rates, and the U.S. dollar index, and also, tying overall demand to industrial production.

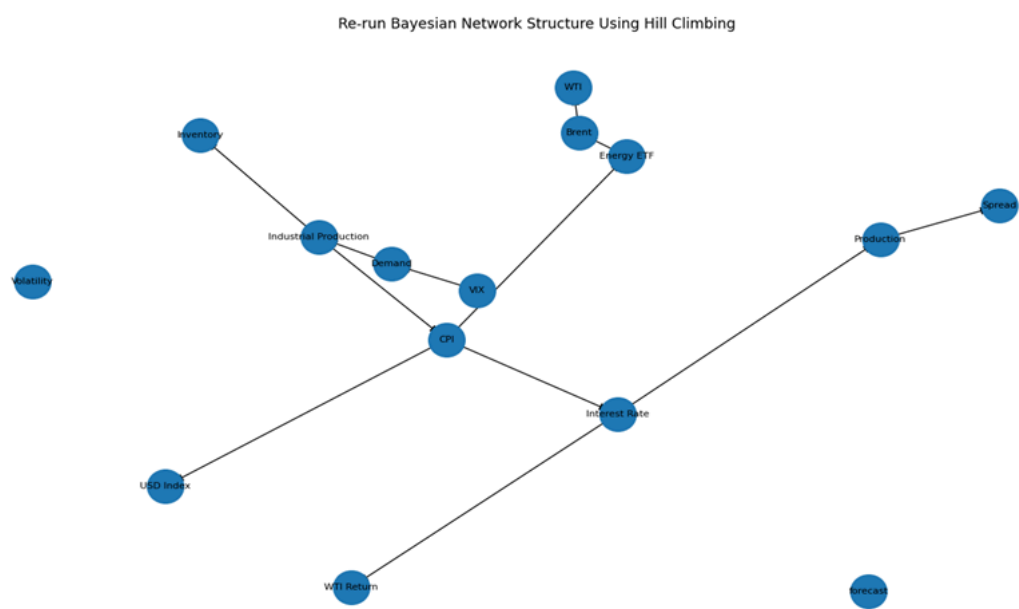


Figure 5. Re-run network structure learned through hill climbing.

The results from the model re-run are shown below. The validation accuracy was 28.57%, and the testing accuracy was 42.86%. The corresponding validation error was 71.43%, and the testing error was 57.14%. We organized our data into market states, rebuilt the network using a hill-climbing algorithm, verified our work with the validation set before the final test, and then, compared our predictions against the reality of the world..

That said, we never expected our final numbers to match Alvi's results. We worked with a different timeframe, data structure, and different definitions of variables utilized.

Table 5. Step 4 Forecasting Results and Alvi Benchmark

Metric	Value
--------	-------

Validation Accuracy (%)	28.5700
Validation Error (%)	71.4300
Testing Accuracy (%)	42.8600
Testing Error (%)	57.1400
Alvi Reported Validation Error (%)	67.8600
Alvi Reported Testing Error (%)	42.8600

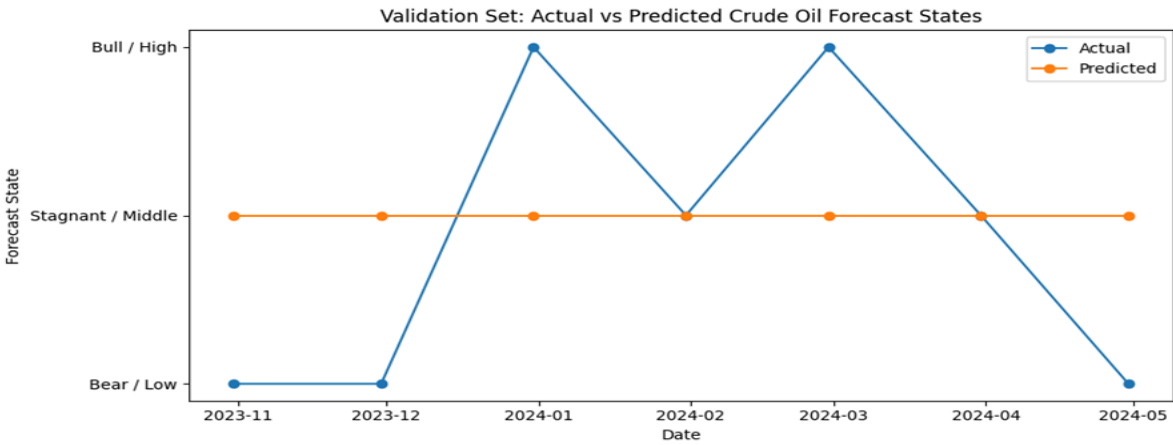


Figure 6. Validation set: actual versus predicted forecast states.

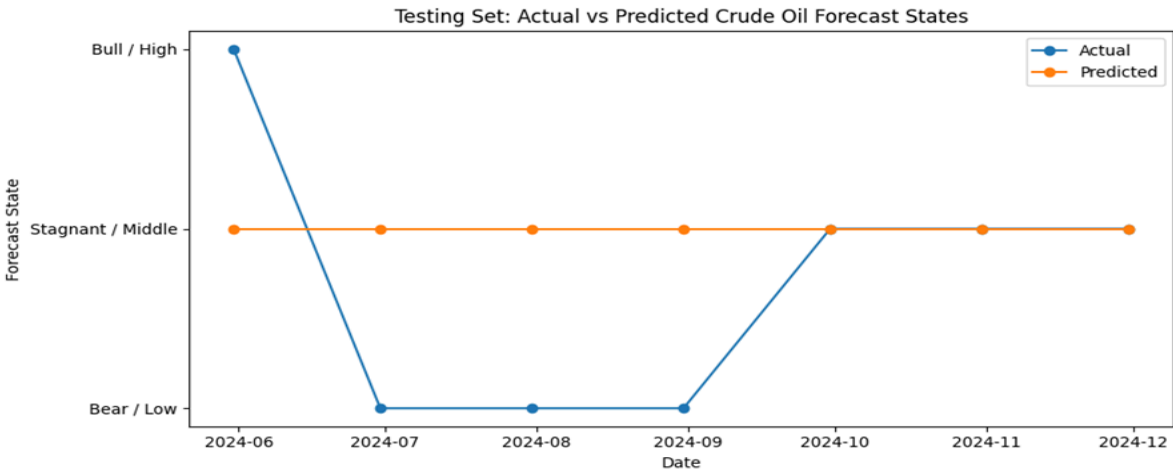


Figure 7. Testing set: actual versus predicted forecast states.

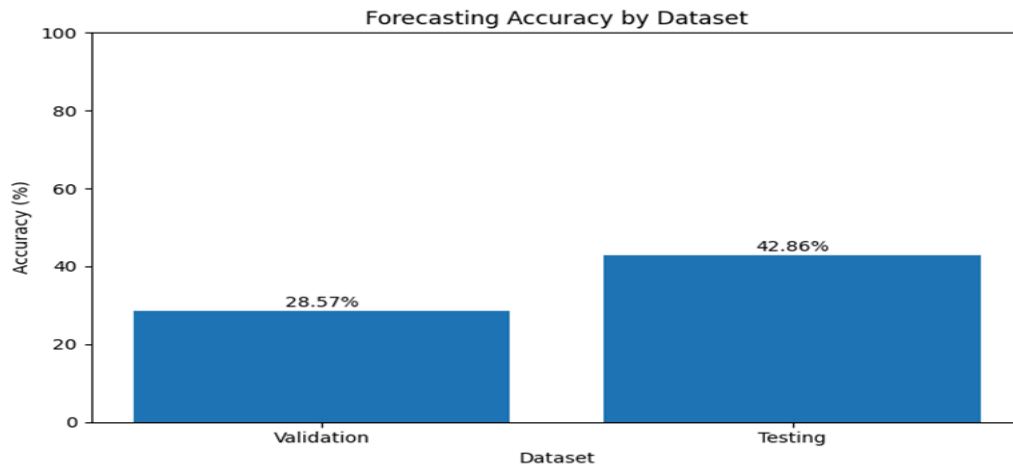


Figure 8. Forecasting accuracy summary.

STEP 5: INTERPRETATION OF THE RESULTS

We were careful when looking at our final testing accuracy of 42.86%. Because of how the next-month target is set up, we ended up with seven months of data in our final test block.

This testing number came from our untouched holdout data, especially when you compare it to the lower 28.57% accuracy. We saw this during validation, but it's still a very small window to judge. To keep things in perspective, remember what this model is trying to do: predicting whether next month's oil market will be weak, neutral, or strong. And so, even when you can't pin down an exact price tag, having that kind of direction is helpful for making smart portfolio and risk management decisions.

Table 6. Testing Accuracy Result

Model	Testing Accuracy (%)	Testing Error (%)
Bayesian Network Forecast	42.8600	57.1400
Naive Majority-State Baseline	42.8600	57.1400

Both achieved 42.86% testing accuracy. The result explains that the current model contains some structure that is not good enough to be an independent rule.

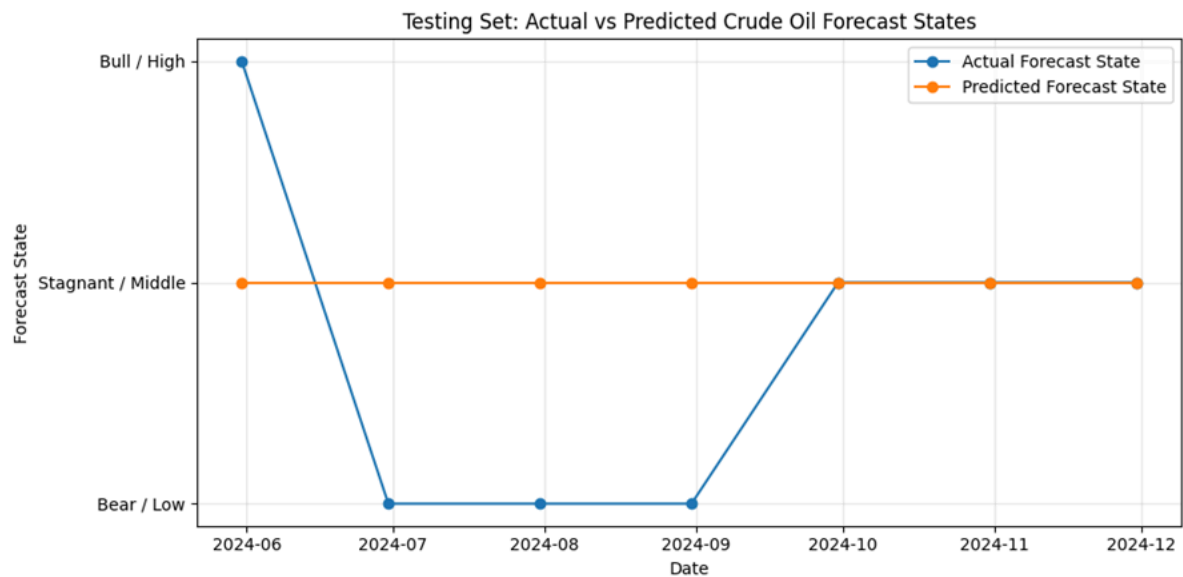


Figure 9. Testing set: actual versus predicted crude oil forecast states.

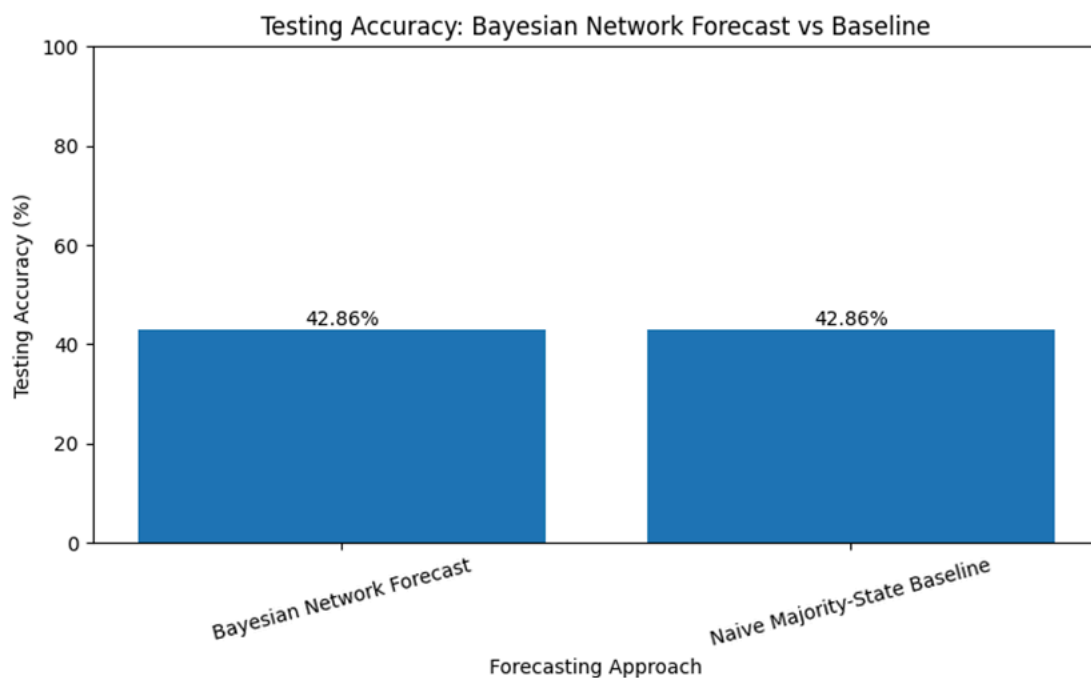


Figure 10. Testing accuracy: forecasting framework versus naive baseline.

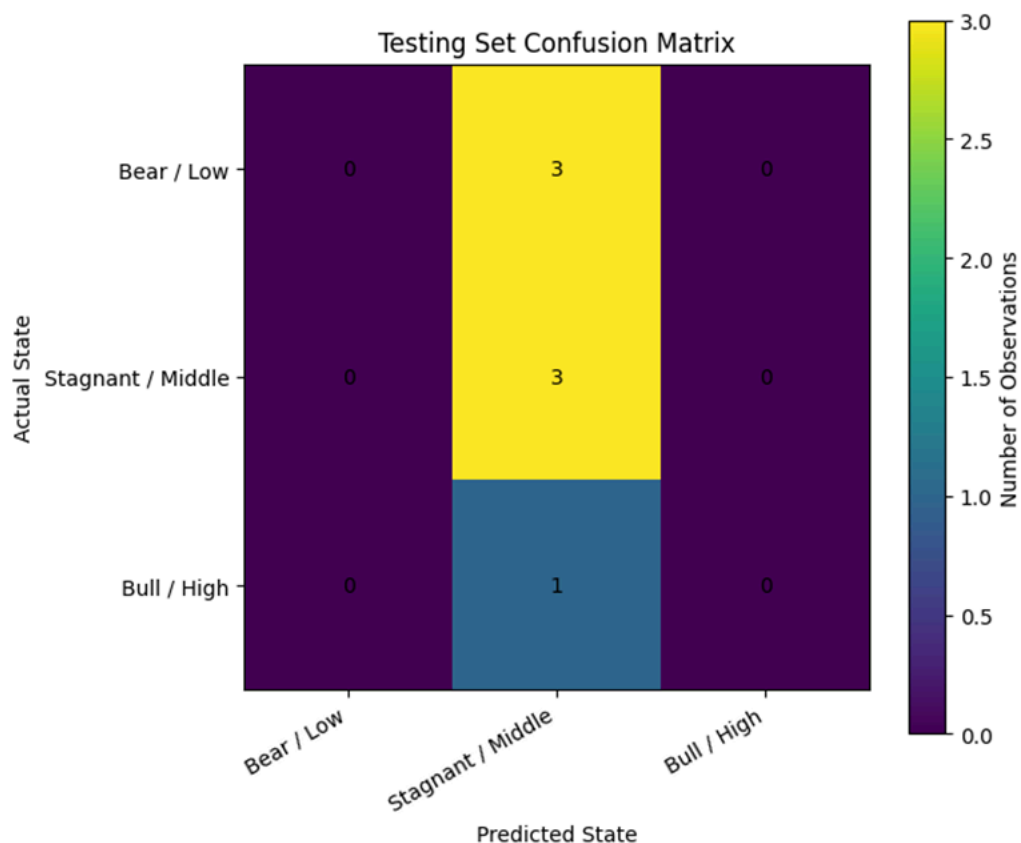


Figure 11. Testing confusion matrix.

Looking at the confusion matrix and the state-level breakdown, it's why our overall accuracy ended up being so under-whelming. During the testing phase for example, the model played it as safe as possible by predicting a "Stagnant/Middle" market for every single observation. Because of this approach, it scored a 100% on the months that were flat and neutral. The obvious downside, though, is that it missed the real market swings, thus scoring a flat 0% for catching the "Bear/Low" drops and another 0% for the "Bull/High" spikes.

Table 7. Testing the State-Level Accuracy

Actual State	Number of Actual Observations	State-Level Accuracy (%)
Bear / Low	3	0.0000
Stagnant / Middle	3	100.0000
Bull / High	1	0.0000

When it comes to forecasting crude oil, stable periods aren't really the focus. For traders and risk managers, the value lies in how the turning points are captured. Unfortunately for us, our model is

just playing it way too safe, that is too cautious. It does a good job of spotting quiet, and neutral conditions, but it misses directional swings. There is a small upside to this: predicting a flat, "middle" state cuts down on wild, and extreme false alarms. But at the end of the day, by sticking to the safe middle ground, the model misses the market moves we need it to find in an accurate way.

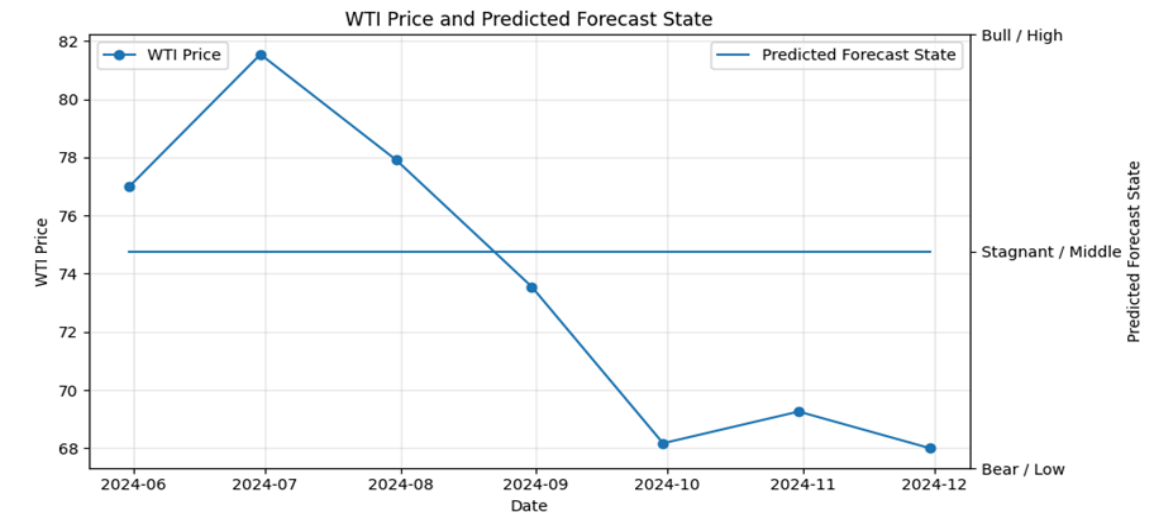


Figure 12. WTI price with predicted forecast state.

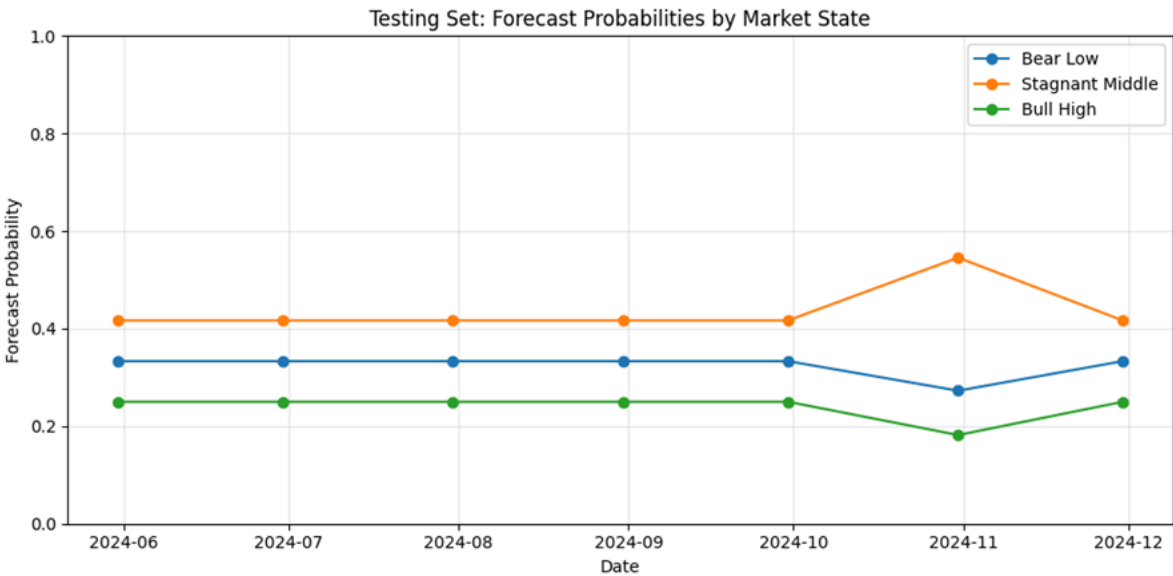


Figure 13. Testing-period forecast probabilities by market state.

The model's confidence was almost always clustered right around that "Stagnant/Middle" option, which makes sense as to why it kept spitting out flat, and neutral forecasts. This insight also gives us a roadmap for how to improve the project going forward. Any future version of this model needs to be recalibrated so it can spot those big directional swings.

STEP 6: Evaluation of the Contributions of Alvi's Dissertation

Danish Alvi's dissertation blended machine learning, computational finance, and macroeconomic theory to forecast commodity markets. Even though it succeeded in creating a framework for organizing crude oil data, our evaluation is that its achievements are a bit of a mixed feeling. We have categorized our assessment into what the research got right, where it fell short, and what still needs to be proven.

Where the Dissertation got it right.

The project is successful in a few specific areas. Arguably, its main contribution is how it handled data inputs. Specifically, it used Hidden Markov Models (HMM) to chop continuous financial data into market states. However, because crude oil moves through different cyclical regimes, the design choice is a good fit. Also, Alvi bridges the gap between complex math and actual execution by utilizing Python-based graphical tools.

Good Concepts the dissertation presented, But Incomplete Execution: Several parts of the paper's goals were partially achieved, mostly because the theory didn't, in detail, cover them well..

Automation: The dissertation aimed to build a fully autonomous decision-making system. Though the structure-learning algorithms reduce manual modeling work, our own project proved that you need to rely on human experts to clean the data, pick the right variables, and ensure the outputs make economic sense, hence it is not entirely automated.

Macro Strategy & Policy: The framework was also designed to act as a systematic global macro strategy and a practical tool for policymakers. Our validation checks, however, showed that the forecasting performance is kind of mixed. Furthermore, the paper never tested the model against a real-world policy situation, which was supposed to prove that it can work in the real world.

The Major Gaps identified (Live Trading and Alpha)

- It argued that we should replace traditional volatility models (like EGARCH-M) with Bayesian-derived views. It's a good idea in theory, but it skips the necessary Black-Litterman portfolio comparison needed to prove it can work.
- Similarly, even though the paper outlined a basic logic for taking long, short, or neutral market positions, it ignored the realities of live trading. You cannot claim to have a working,

alpha-generating trading mechanism without accounting for transaction costs, execution slippage, and other important factors..

Ultimately, Alvi's dissertation is a brilliant methodological prototype. It proves that modern oil forecasting needs to look beyond price charts and incorporate broad macroeconomic and physical-market data.

Step 7: DISCUSSION AND PRACTICAL APPLICATION.

This model is not an automated trading bot. It scored 42.86% accuracy rate in our final tests. That ties with a simple, and educated guess. If you let it pull the trigger on trades all by itself, you are asking for inaccurate predictions. It didn't pass a basic rule of thumb, so it always needs human supervision.

So, think of it as a gut-check tool for portfolio managers. It can act as an extra layer of screening to help you position your trades. Below is what that can look like in the real world application.

When the model signals a strong market: Look at the physical evidence. Are inventories dropping? Is consumer demand holding steady? If the real-world data agrees with the model, it is a green light to increase your energy investments.

When it warns of weakness: Check the market volatility. If things are getting choppy, take the hint. It is time to play defense, tighten your risk limits, or start hedging.

When it is neutral: There is a need to be patient. The data shows that sitting on your hands is usually a lot smarter than making a risky bet in a flat market.

The big trap you can fall into is overconfidence. Our test sample was small, meaning, just a few weird months could easily swing that accuracy percentage around. Always treat this framework as a cost-effective early-warning system. It organized noisy global data into a clean snapshot, but it is meant to help human judgment, never to replace it. Going forward, our next goal is to tweak the system so it gets much better at catching those acute market turns, because that is where the real value is won or lost.

List of References

1. Alvi, Danish A. Application of Probabilistic Graphical Models in Forecasting Crude Oil Price. University College London, 2018. arXiv:1804.10869.
2. Board of Governors of the Federal Reserve System and Federal Reserve Bank of St. Louis. Federal Reserve Economic Data (FRED)
3. Cboe Global Markets. Cboe Volatility Index (VIX).
4. Group Work Project 1 Report. MScFE 660 Risk Management, WorldQuant University, 2026. Internal coursework report.

5. Group Work Project 2 Report. MScFE 660 Risk Management, WorldQuant University, 2026. Internal coursework report.
6. Koller, Daphne, and Nir Friedman. Probabilistic Graphical Models: Principles and Techniques. MIT Press, 2009.
7. Pearl, Judea. Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference. Morgan Kaufmann, 1988.
8. Scutari, Marco. "Learning Bayesian Networks with the bnlearn R Package." Journal of Statistical Software, vol. 35, no. 3, 2010, pp. 1-22.
9. U.S. Energy Information Administration. Open Data and Petroleum Data Series.
10. Yahoo Finance. Historical Market Data for WTI, Brent, VIX, and Energy ETF Series.